

A Continuous Schur Multiplier with No Continuous Hilbert Factorization

Automated Math Discovery Candidate Counterexample

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Status. Candidate full counterexample, likely valid pending expert review. It gives a negative answer to Steen–Todorov–Turowska Question 4.13(i), as predicted in the target paper.

Source Question

Steen, Todorov, and Turowska ask in Question 4.13(i) of [2] whether every continuous Schur multiplier $\varphi : X \times X \rightarrow \mathbb{C}$ on a locally compact space with a regular Borel measure has a representation

$$\varphi(x, y) = \langle a(x), b(y) \rangle_{\ell^2}$$

with continuous bounded functions $a, b : X \rightarrow \ell^2$. Arhancet [1] proves the positive-kernel version and writes that the answer to part (i) should be negative. The example below proves that prediction, even for a compact metric space with a full-support probability measure.

It was shown in [16] that if a continuous function φ of Toeplitz type defined on the direct product $G \times G$, where G is a locally compact group, is a Schur multiplier, then the functions $a, b : G \rightarrow \ell^2$ in the representation $\varphi(x, y) = \langle a(x), b(y) \rangle_{\ell^2}$, can be chosen to be continuous. It is thus natural to ask the following questions:

Question 4.13. *Let X be a locally compact topological space equipped with a regular Borel measure. Suppose that $\varphi : X \times X \rightarrow \mathbb{C}$ is a continuous Schur multiplier.*

(i) *Do there exist continuous bounded functions $a, b : X \rightarrow \ell^2$ such that $\varphi(x, y) = \langle a(x), b(y) \rangle_{\ell^2}$ for all $x, y \in X$?*

(ii) *If φ is moreover positive, can one choose a continuous bounded function $a : X \rightarrow \ell^2$ such that $\varphi(x, y) = \langle a(x), a(y) \rangle_{\ell^2}$ for all $x, y \in X$?*

(iii) *Assuming that φ is a local (resp. positive local) Schur multiplier, can a similar choice be made with a and b (resp. a) not necessarily bounded?*

Proof Intuition

Take infinitely many finite Schur multipliers, each of norm one, but make their scalar entries tend uniformly to zero. The resulting block diagonal kernel is continuous after all blocks are joined at one point at infinity. If continuous Hilbert-space factors existed, subtracting their values at infinity

would make both factors uniformly small on late blocks. Their product would then force the Schur multiplier norms of those blocks to tend to zero, contradicting the norm-one construction.

Normalized Hadamard matrices achieve the needed separation: their entries are $n^{-1/2}$, while their Schur multiplier norms are exactly one.

Main Result

Theorem 1. *There exist a compact metrizable space X , a regular Borel probability measure μ with full support on X , and a continuous function $\varphi : X \times X \rightarrow \mathbb{C}$ such that:*

1. φ induces a Schur multiplier

$$M_\varphi : \mathcal{B}(L^2(X, \mu)) \longrightarrow \mathcal{B}(L^2(X, \mu))$$

of norm one;

2. *there are no continuous functions $a, b : X \rightarrow \ell^2$ satisfying*

$$\varphi(x, y) = \langle a(x), b(y) \rangle \quad (x, y \in X).$$

In particular, Question 4.13(i) of [2] has a negative answer. Since X is compact, the word “bounded” in that question is automatic for continuous a and b .

We first record the elementary factorization estimate that drives the obstruction.

Lemma 1 (Factorization bound). *Let E be a finite set and let $\psi = (\psi_{ij})_{i,j \in E}$. If $\psi_{ij} = \langle \xi_i, \eta_j \rangle$ for vectors in a Hilbert space, then the Schur multiplier S_ψ on $\mathcal{B}(\ell^2(E))$ satisfies*

$$\|S_\psi\| \leq \left(\sup_{i \in E} \|\xi_i\| \right) \left(\sup_{j \in E} \|\eta_j\| \right).$$

Proof. Define $Ue_i = e_i \otimes \xi_i$ and $Ve_j = e_j \otimes \eta_j$. Up to the irrelevant choice of inner-product convention,

$$S_\psi(T) = U^*(T \otimes I)V.$$

Hence $\|S_\psi\| \leq \|U\| \|V\|$, which is the displayed bound. □

Proof of Theorem 1. Let

$$X = \mathbb{N} \cup \{\infty\}$$

be the one-point compactification of the discrete natural numbers. Thus $n \rightarrow \infty$ in X as $n \rightarrow \infty$ in the usual sense, and X is homeomorphic to $\{0\} \cup \{1/n : n \geq 1\}$. Put

$$\mu(\{n\}) = 2^{-n} \quad (n \geq 1), \quad \mu(\{\infty\}) = 0.$$

This is a regular Borel probability measure. It has full support: every finite point is a positive-mass open atom, and every neighborhood of ∞ contains a cofinite tail of positive mass.

Partition $\mathbb{N}$ into successive intervals

$$I_k = \{2^k - 1, 2^k, \dots, 2^{k+1} - 2\}, \quad |I_k| = n_k := 2^k, \quad k \geq 1.$$

Let H_k be the Sylvester Hadamard matrix of order n_k . Thus all its entries are ± 1 and

$$H_k H_k^* = n_k I_{n_k}.$$

After identifying the rows and columns with I_k , define

$$\varphi(i, j) = \begin{cases} n_k^{-1/2} (H_k)_{ij}, & i, j \in I_k \text{ for the same } k, \\ 0, & \text{otherwise,} \end{cases} \quad \varphi(\infty, x) = \varphi(x, \infty) = 0. \quad (1)$$

We check continuity. Finite points are isolated. At (∞, j) , where j is finite, the kernel is identically zero once the first coordinate is outside the one finite block containing j ; similarly at (i, ∞) . At (∞, ∞) , points in distinct blocks give zero, while points in a common block I_k have absolute value $n_k^{-1/2} = 2^{-k/2} \rightarrow 0$. This is exactly the required continuity.

Next we compute the Schur norm. In the Hilbert direct sum $\mathcal{H} = \bigoplus_{k \geq 1} \ell^2(I_k)$, let ξ_i be the row of $n_k^{-1/2} H_k$ corresponding to $i \in I_k$, placed in the k th summand, and let η_j be the standard basis vector corresponding to $j \in I_k$ in that same summand. Then

$$\|\xi_i\| = \|\eta_j\| = 1, \quad \varphi(i, j) = \langle \xi_i, \eta_j \rangle.$$

The same operator factorization as in Lemma 1 gives $\|M_\varphi\| \leq 1$. This applies on $L^2(X, \mu)$ because the normalized atom indicators identify it unitarily with $\ell^2(\mathbb{N})$, and multiplication of integral kernels by φ becomes entrywise matrix multiplication; the atom weights cancel in this normalized basis.

The reverse inequality is already visible on every block. Write $\varphi_k = n_k^{-1/2} H_k$. Entrywise multiplication gives

$$S_{\varphi_k}(H_k) = n_k^{-1/2} J_{n_k},$$

where J_{n_k} is the all-ones matrix. Since

$$\|H_k\| = \sqrt{n_k}, \quad \left\| n_k^{-1/2} J_{n_k} \right\| = \sqrt{n_k},$$

we get $\|S_{\varphi_k}\| \geq 1$. Compression to the block gives $\|M_\varphi\| \geq 1$, and hence $\|M_\varphi\| = 1$.

Suppose, toward a contradiction, that continuous $a, b : X \rightarrow \ell^2$ satisfy $\varphi(x, y) = \langle a(x), b(y) \rangle$ everywhere. Set

$$\tilde{a}(x) = a(x) - a(\infty), \quad \tilde{b}(y) = b(y) - b(\infty).$$

The three identities

$$\langle a(\infty), b(y) \rangle = \varphi(\infty, y) = 0, \quad \langle a(x), b(\infty) \rangle = \varphi(x, \infty) = 0, \quad \langle a(\infty), b(\infty) \rangle = 0$$

show that the centered maps still factor the same kernel:

$$\varphi(x, y) = \langle \tilde{a}(x), \tilde{b}(y) \rangle.$$

Continuity at ∞ now implies

$$A_k := \sup_{i \in I_k} \|\tilde{a}(i)\| \longrightarrow 0, \quad B_k := \sup_{j \in I_k} \|\tilde{b}(j)\| \longrightarrow 0.$$

Applying Lemma 1 to the restriction to $I_k \times I_k$ would give

$$1 = \|S_{\varphi_k}\| \leq A_k B_k \longrightarrow 0,$$

an impossibility. Thus no continuous Hilbert factorization exists. $\square$

Scope and Novelty

The counterexample is not positive; this is necessary because the positive case was proved affirmatively in [1]. It uses only a compact convergent sequence, a full-support probability measure, and real-valued kernels. Bounded searches through July 22, 2026 found the original question, Arhancet's negative prediction, and general factorization literature, but no published counterexample of this form. Novelty confidence is therefore moderate pending a specialist literature review.

References

- [1] C. Arhancet, *Dilations of Markovian semigroups of measurable Schur multipliers*, Canad. J. Math. 74 (2022), 59–93; arXiv:1910.14434.
- [2] N. M. Steen, I. G. Todorov, and L. Turowska, *Local operator multipliers and positivity*, J. Funct. Anal. 267 (2014), 80–111; arXiv:1206.3889.